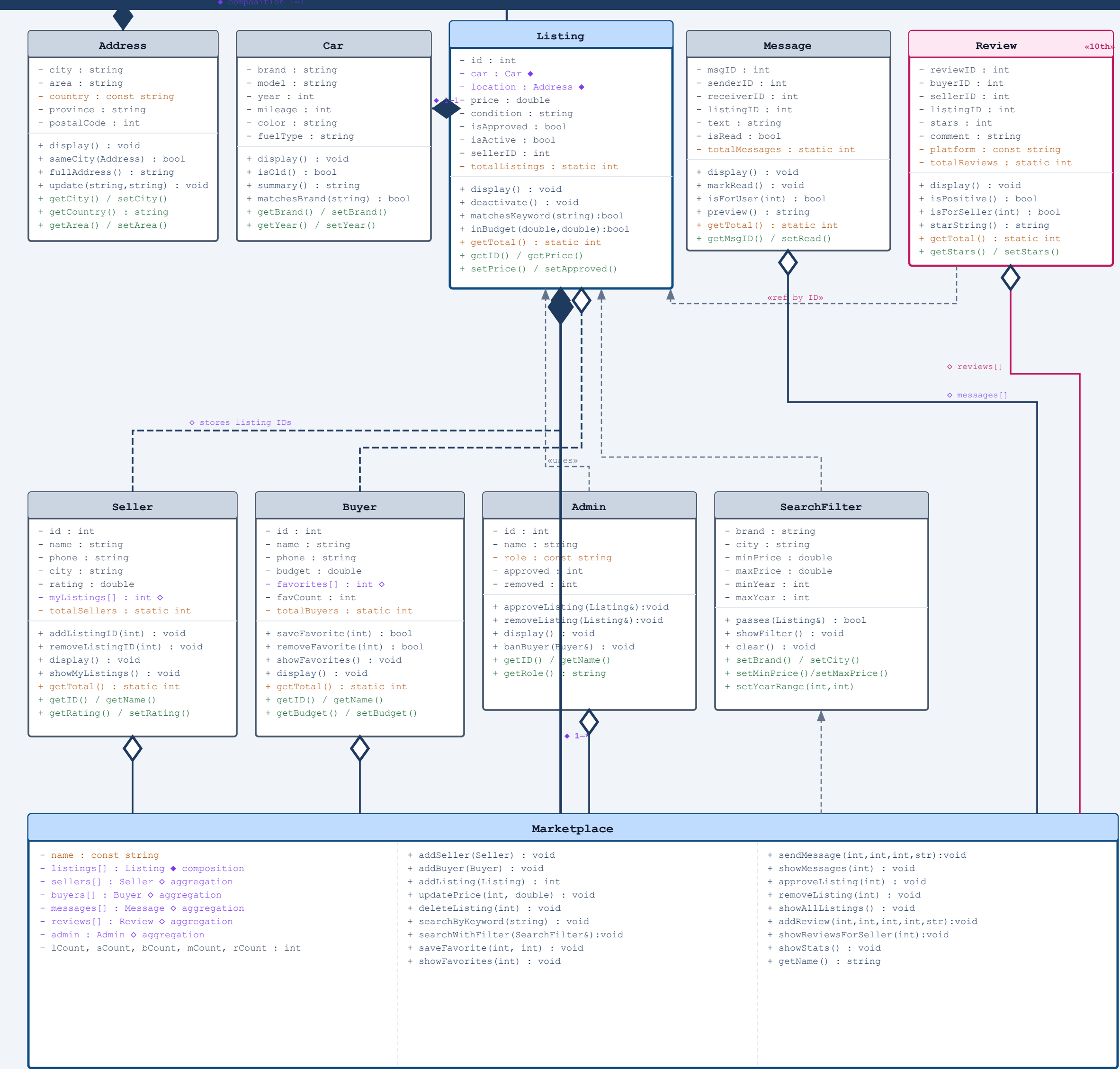
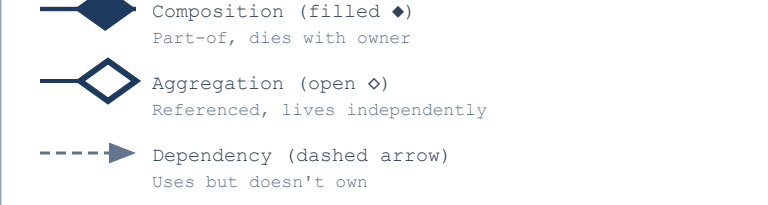


CAR MARKETPLACE SYSTEM – UML Class Diagram (10 Classes)



LEGEND & NOTES

Relationship Symbols:



Colour Key:

- Orange = parent / state's member

- Orange = const / static member
- Purple = relationship field
- Dark blue = methods
- Green = getters / setters
- Pink = Review (10th class)

```
Composition (owns, 1-1):
```

- Listing $\blacklozenge$ Car
- Listing $\blacklozenge$ Address
- Marketplace $\blacklozenge$ listings[] (1-*)

Aggregation (independent lifetime):

- Marketplace $\diamond$ — sellers[]
- Marketplace $\diamond$ — buyers[]
- Marketplace $\diamond$ — messages[]
- Marketplace $\diamond$ — reviews[]
- Marketplace $\diamond$ — admin
- Seller / Buyer $\diamond$ — listing IDs

```
Static Members (class-level):
```

- Listing::totalListings + getTotal()
- Seller::totalSellers + getTotal()
- Buyer::totalBuyers + getTotal()
- Message::totalMessages + getTotal()
- Review::totalReviews + getTotal()

Total Classes: 10 ✓